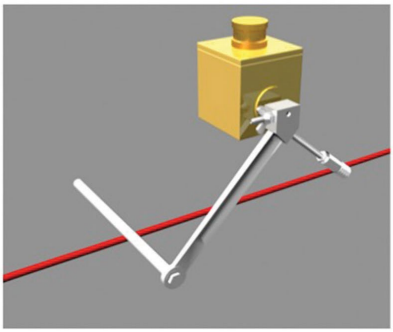


The CS200 is an elevation control sensor and can be used with a profile ski. The sensor should be set up and nulled/surveyed with the wand arm at a 45° angle trailing the sensor with the connector facing forward. The contact sensor is unaffected by temperature variations and measures the changes in the elevation of a control reference which could be a ski, curb, existing mat or existing grade by measuring the change in angle of the wand arm.



The CS200 rotary contact sensor can also be used with a string line wand to reference an object. When mounted correctly the wand arm will be facing to the rear and will be at a 45º angle with the wand centered on the string. The wand pressure can be adjusted and configured to reference above or below a string-line. Once the contact sensor has been nulled/benched, any change in the elevation or wand angle will be sent to the display. The display evaluates the information and sends a signal to the ECM to raise or lower the tow point cylinder.



The AS200 angle sensor is a sensor that functions as a precision carpenter's level. The angle sensor reads the slope of the screed and sends an electronic signal to the display. The angle sensor contains an electronic device that will constantly measure the slope of the screed. The AS200 angle sensor is mounted on the transverse beam of the screed.

The sensor is calibrated by editing the slope value in the measured values area. This enables the operator to maintain a desired positive or negative slope from ±10° (±17.6%) but is limited by machine capabilities.



Because the tractor display is a full touch screen display, the operator MUST push the Display unlock button (1) for the touch function to work. The LED light must be lit for this operation. At that time the touch functions will be allowed. The operator can then access the Grade and Slope system interface by touching the Grade and Slope Icon (2) from any screen that shows the icon.

NOTE: After three minutes the display will automatically lock again.



The screed display is a combination soft key button and/or touch screen. The operator will need to press and release the Grade and Slope soft key button from any screen that shows the icon to enter the grade and slope system interface.





There are two different types of displays on the F-series paver that serve as the operator interface for the Cat Grade and Slope system. The operator on the tractor has a total touch screen display that has no external buttons. This display does not have any internal or external light bars for grade indication. The screed operator display has a combination touch and soft key button control interface with internal and external light bars for grade indication. These displays will perform the same functions but the operator interface will be slightly different on each.

NOTE:

- **Tractor displays only allow dual screen mode (shown).**
- **Screed displays can be set to dual screen mode or single screen mode (shown).**



Once in the Grade and Slope operating interface on the tractor, the operator will see in the lower left corner, a padlock icon (1). This icon is used to: unlock only the icons on the sides (2) of the display, go to the previous screen, or “okay” a new setting or change made. The unlock function (1) will only work for three seconds or one icon button touch. The system operates in this way so no “unwanted” change happens unless the operator requests it. Area (3) is allowed without having to touch the Lock icon (1) but when a value is changed you will have to “okay” the change and that requires the use of the Lock icon (1).

NOTE: **The Tractor display is a full touch screen so remember, the screen must be unlocked first before the touch function works as explained in slide three and after three minutes the display will automatically lock again.**



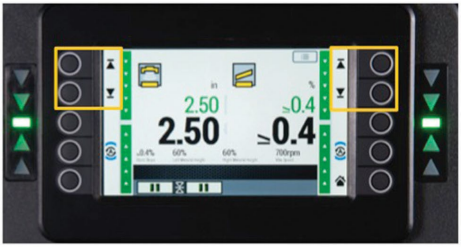
The screed displays are a combination touch screen and/or soft key function. In area 1, the operator can only use the soft key buttons to utilize those functional icons. In area 2, the operator can use the touch screen function or a combination of soft key button and touch screen. Remember for touch screen operation, the screen must be activated with the display unlock button on the key pad (3).

NOTE: The menu structure is the same with either display, but the operator interface is different between the two. This presentation will use the Screed Display to explain the operation of the system.



The middle section of the screen is called the Guidance Screen. The Guidance screens display a mix of text and graphics that provide information such as the slope and elevation or different menus. There are many different sections to this screen that will be explained within this presentation.

NOTE: Dual screen mode shown.



The soft keys adjacent to the arrow icons can be used in different ways. When an operator wants to use these arrow keys to move the tow point cylinders, they must first actuate the machine logic to allow for movement.

To accomplish this, the operator will need to place the machine in the active mode or meet the **"Paving Parameters"** which are:

1. Engine Running
2. Place the Propulsion Mode Switch in the **Pave Mode**.
3. Place the screed into **float**.
4. Parking Brake **"On"**.
5. **Squeeze propulsion trigger and stroke propulsion handle forward.**

Once the “Paving Parameters” are met, the operator can press the arrow soft key buttons and manually raise or lower the tow point cylinders to the required position.

In the Automatic mode, the operator can change the “target” value for elevation or slope.

When in the Configuration Mode the operator has the ability to select a desired menu item and change the values.

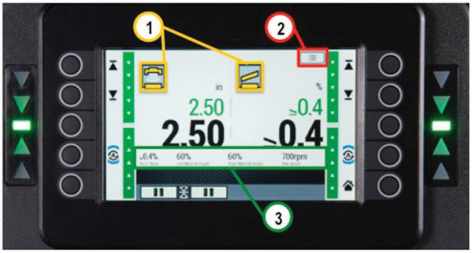
NOTE: While paving, the Paving Parameters must also incorporate the “Target speed above zero” to allow for automatic operation.



Item 1 is the Bench soft key. This function is only present in the manual mode, the icon will disappear when placed in the Automatic mode. Press and hold the soft key for three seconds to bench the position or height of the grade sensor to the reference. When in the slope mode, this will bench the position or angle of the screed.

Item 2 is the Automatic / Manual mode soft key. This function will place the system in the Automatic control mode (control of tow point cylinders) or in the Manual mode (no control of tow point cylinders) when pressed and released.

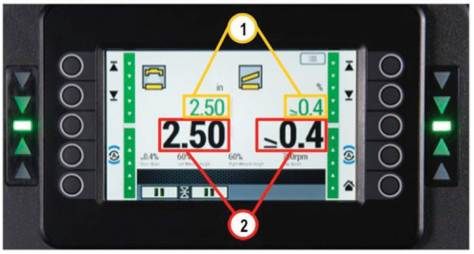
Item 3 is the Home or Machine Health soft key. When pressed and released the system will be taken out of the Grade and Slope system screens. If the operator is in the automatic mode, the system will continue to control the tow points at the benched position.



Item 1 is the Sensor Select icon. This is a touch screen item that allows the choosing of an available sensor that is attached to the machine. Icons will appear with the available plugged in sensors.

Item 2 is the Menu/Operator setting icon. This icon is also a touch function. This menu allows the operator to configure or customize the grade control system, to use system diagnostics and/or to calibrate the system.

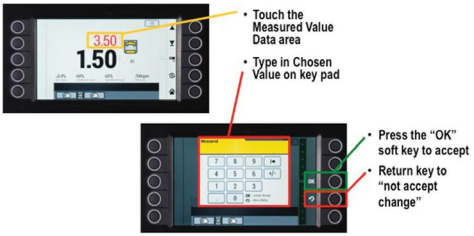
The last item, 3, is the text slider bar. The text slider bar allows the operator to view configurable information about different machine and screed system settings while in the Grade and Slope system screen. If the operator touches the text slider bar with one finger and slides it to one side or the other they would be able to view several pieces of information.



The Measured Value area (1) shows the actual sensor data in real time. When the sensors are configured and/or calibrated and the screed rises or falls the actual sensor data that it sees will be displayed in this area.

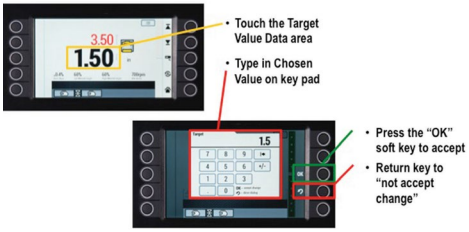
In the Target Value area (2), the values displayed are the targeted elevation or the targeted slope of the screed.

The configured or calibrated numbers for these areas can be adjusted by following the instructions on the next slide or page.



The operator can edit the value in the measure values data area by touching the screen. A key pad will appear on the screen. The operator touches the value they wish to display in this area. Once complete, the "OK" soft key must then be pressed to accept the number chosen. It is common practice to use the value that is measured while paving whether in the grade mode or the slope mode. If editing the number in the slope mode, the correct angle direction must be chosen also.

NOTE: If this action is done in the manual mode, the system must be re-benched. If it is done in the automatic mode, the number will automatically be pushed to the target value data area.



The "Target" is the position you want the screed to be at or drive to. If the target data area on the display is touched a key pad will appear. The operator can choose a new target. In the Automatic Mode if a new target value has been added and the "OK" soft key is pressed the system will automatically drive the tow point to the new target position if the paving parameters have been met, so care must be taken when doing this operation.

In the manual mode the number will change but the system won't drive to the new target unless the operator goes into the automatic mode and has met the paving parameters.



There are two different types of indicators on this Grade control system. There is the internal vertical graph light bars which are located on the touch screen and external bars that are located on the outboard side of the display. The operator can configure these light bars to work in the automatic mode only, in the automatic/manual mode, or turned off. The external light bars have two different color LED's, amber or green and the internal is either red or green. The flashing pattern of these indicators change depending on the state of the grade control correction. For a detailed explanation on the indicators refer to the owners manual for this system. The brightness of these indicators will follow the screen brightness setting when adjusted.



On the bottom lower banner of the Guidance screen there is a block that shows the status of the grade control system. This lower banner will be present no matter what view the operator is looking at whether it is the grade control screen or one of the machine health screens. In the picture shown, the Paver is in the Automatic mode on both sides of the machine and the machine is moving (Paving Parameters met). When the system has both sides set to the manual mode, two manual mode (hand) icons will be shown. If the paver operator has the propulsion lever in neutral with the grade control in the Automatic mode, the “Pause” icons will blink alternately with the “Auto” icons. When one side of the paver is in Automatic mode and one is in the manual mode, the side activated in automatic will have the “Auto” icon and the other side will show the Manual mode icon.





Preparing to Pave (Grade)

First, set up the paver using Paving by the Numbers. Then choose the proper sensor type by touching the Sensor Select Icon. Once the proper sensor is selected press “Okay” soft key. The sensor is now chosen.

NOTE: Only the sensors that are plugged into the machine will be available for use.



Preparing to Pave (Grade) ***(Continued)***

The position of the sensor along the tow arm determines how fast the screed will react to a change. By positioning the sensor, we can build profile or rideability. We need to evaluate job specifications and grade related conditions to determine the desired position that will produce the required results.

Automation sensors will always move the tow point up or down to maintain the sensors bench point or null point. In effect, the location of the sensor becomes our only control point. The tow point becomes only a point from which we raise or lower the tow arm to reestablish the bench point or null point at the sensor. The effective length of tow arm becomes the distance between the control point (sensor) and the pivot point (trailing edge of the screed).



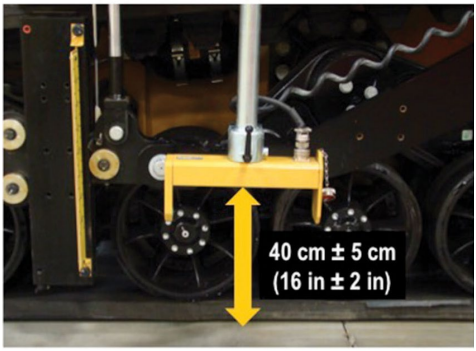
Because the screed must travel five tow arm lengths to fully react to a correction, shortening the effective tow arm length shortens the distance the paver must travel to complete a correction. Positioning the sensor to the rear or inline with the auger shaft produces a quick change (or reaction) of the screed. This placement will have a 4 to 1 ratio. Meaning that for every 2.5 cm (1 in) of sensor movement up or down, the tow point will move up or down 10 cm (4 in). This position is used to match existing surfaces or a curb line.

Rule: Sensor(s) positioned inline with the augers builds PROFILE.



Positioning the sensor forward or near the tow point causes gradual changes in tow point movement. Again, because the screed must travel five tow arm lengths to fully react to a correction, increasing the effective tow arm length increases the distance the paver must travel to complete a correction. Positioning the sensor near the tow point decreases reaction time of the screed. This placement will have a 1 to 1 ratio. Meaning that for every 2.5 cm (1 in) of sensor movement up or down, the tow point will also move up or down 2.5 cm (1 in). This position is used to build ridability or smoothness.

Rule: Sensor(s) positioned near the tow point or 1/4 of the tow arm length back from the tow point builds RIDABILITY.

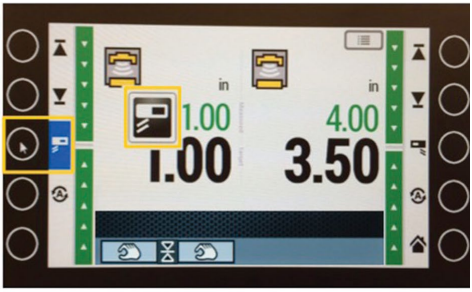


Set the height of the grade sensor to 40.6 cm +/- 5 cm (16 +/- 2 in) or the angle of the wand arm to a 45º angle on the contact sensor above the reference.

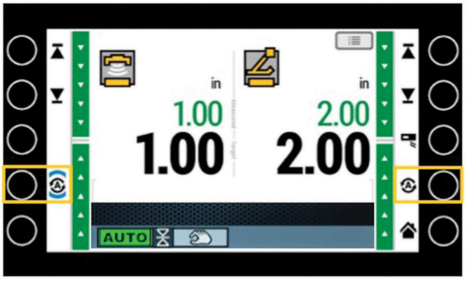




Now adjust the number in the Measured Value Data area to display the actual paving depth that will be paved. To do this, touch the “unlocked” display Measured Value Data area on the Guidance screen. A key pad will appear on the screen. Touch the actual paving depth numbers on the screen and press the “OK” soft key to enter the value.



The sensor position height can now be benched. Press and hold the bench soft key until you hear an audible beep or until the bench icon appears on the guidance screen. This will tell the control system to hold the screeds current elevation position. Do this on both sides. This will also move the Measured data number that was entered to the Target value data position.



Press and release the Manual / Automatic soft key to place the system into the automatic mode. Repeat on the other side.

Once paving begins, the control system will adjust the tow point cylinders to maintain the selected reference point. If any adjustment needs to be made the operator can press the Arrow soft key the direction needed to maintain specification.